

Chain Rule 03: Scalar Cosine with Radical Argument

Course: Mathematics-I (Differential Calculus)

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প্রদত্ত সমীকরণ: $y = 8 \cos \sqrt{x}$

নির্ণেয়: $\frac{dy}{dx}$

ENGLISH NOTE — SCALAR-TRIGONOMETRIC-RADICAL CHAIN RULE FORMULATION

The problem combines a scalar constant multiplier 8, a cosine envelope, and a square root inner function:

$$y = 8f(g(x)), \quad \text{where } f(u) = \cos u, \quad u = g(x) = \sqrt{x}$$

Applying Leibniz's Chain Rule:

$$\frac{dy}{dx} = 8 \cdot \frac{d}{du}(\cos u) \cdot \frac{du}{dx} = 8 \cdot (-\sin u) \cdot \frac{1}{2\sqrt{x}} = -\frac{4 \sin \sqrt{x}}{\sqrt{x}}$$

১. গাণিতিক কাঠামো ও প্রয়োজনীয় সূত্রসমূহ

উপাদান (Component)	ফাংশন রূপ	প্রমিত সূত্র (Standard Formula)
স্কেলার গুণক (Scalar)	8	গুণক গুণক বাইরে অপরিবর্তিত থাকে
বহিঃস্থ ফাংশন (Outer)	$\cos u$	$\frac{d}{du}(\cos u) = -\sin u$
অভ্যন্তরীণ ফাংশন (Inner)	$u = \sqrt{x}$	$\frac{du}{dx} = \frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}}$

ENGLISH NOTE — CRITICAL AMIE EXAM TRAPS & PITFALLS

1. **Negative Sign in Cosine:** Always recall $\frac{d}{dx}(\cos \theta) = -\sin \theta$. Omitting the minus is an automatic mark loss.2. **Inner Radical Derivative:** Multiplying by $\frac{1}{2\sqrt{x}}$ is mandatory.3. **Argument Distinction:** $-\frac{4 \sin \sqrt{x}}{\sqrt{x}} \neq -4 \sin(1)$. Never cancel the argument radical with the denominator radical!

২. ধাপে ধাপে পূর্ণাঙ্গ সমাধান (Step-by-Step Solution)

ধাপ ১: উভয় পক্ষে x -এর সাপেক্ষে অন্তরীকরণ অপারেটর প্রয়োগ করে:

$$\frac{dy}{dx} = \frac{d}{dx}(8 \cos \sqrt{x}) = 8 \cdot \frac{d}{dx}(\cos \sqrt{x})$$

ধাপ ২: শৃঙ্খল নিয়ম (Chain Rule) অনুসারে সাজানো:

$$\frac{dy}{dx} = 8 \cdot (-\sin \sqrt{x}) \cdot \frac{d}{dx}(\sqrt{x})$$

ধাপ ৩: অভ্যন্তরীণ পদ $\sqrt{x}$ -এর অন্তরজ $\frac{1}{2\sqrt{x}}$ বসিয়ে:

$$\frac{dy}{dx} = 8 \cdot (-\sin \sqrt{x}) \cdot \frac{1}{2\sqrt{x}} = -\frac{4 \sin \sqrt{x}}{\sqrt{x}}$$

ধাপ ৪: ধ্রুবক সহগ সরলীকরণ ($8/2 = 4$) করে চূড়ান্ত প্রমিত রূপ:

$$\frac{dy}{dx} = -\frac{4 \sin \sqrt{x}}{\sqrt{x}}$$

চূড়ান্ত উত্তর (FINAL STANDARD ANSWER)

$$\frac{dy}{dx} = -\frac{4 \sin \sqrt{x}}{\sqrt{x}}$$

🔬 ENGLISH NOTE — DOMAIN & RIGHT-HAND DIFFERENTIABILITY

- **Domain of $y(x)$:** Real square roots require non-negative radicands: $x \geq 0 \implies \text{Dom}(y) = [0, \infty)$.
- **Domain of Differentiability ($\frac{dy}{dx}$):** Division by $\sqrt{x}$ excludes $x = 0$: $\text{Dom}\left(\frac{dy}{dx}\right) = (0, \infty)$.
- **Limiting Behavior at the Origin:**

$$\lim_{x \rightarrow 0^+} \frac{dy}{dx} = \lim_{x \rightarrow 0^+} \left(-4 \cdot \frac{\sin \sqrt{x}}{\sqrt{x}} \right) = -4 \cdot 1 = -4$$

💡 ENGLISH NOTE — FORMAL SUBSTITUTION MODEL

Let $u = \sqrt{x} \implies \frac{du}{dx} = \frac{1}{2\sqrt{x}}$.

$$y = 8 \cos u \implies \frac{dy}{du} = -8 \sin u$$

$$\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx} = (-8 \sin u) \cdot \frac{1}{2\sqrt{x}} = -\frac{4 \sin \sqrt{x}}{\sqrt{x}}$$

This 2-step substitution is textbook-standard for AMIE engineering exams.

🎯 AMIE EXAM MARKS DISTRIBUTION STRATEGY

Exam Evaluation Step	Required Mathematical Rigor	Marks
Chain Rule Formulation	Writing $8(-\sin \sqrt{x}) \cdot \frac{d}{dx}(\sqrt{x})$	2.0
Radical Differentiation	Accurate evaluation $\frac{d}{dx}(\sqrt{x}) = \frac{1}{2\sqrt{x}}$	1.5
Scalar Simplification	Simplifying $-\frac{8}{2} = -4$ and final format	1.5
Total AMIE Section-A Marks		5.0 / 5.0